

1. Vector Database Schema (ChromaDB)

The text-based knowledge base is stored in ChromaDB using a flat schema consisting of vector embeddings and associated metadata.

- **Collection Name:** chroma_db_store.
- **Embedding Model:** sentence-transformers/all-MiniLM-L6-v2.
- **Metadata Fields:**
 - filename : The source PDF file name (e.g., Samsung_Washer.pdf).
 - category : Hardcoded as "User Manual".
 - page_content : The raw text chunk (approx. 1000 characters).

2. Image Knowledge Base Schema

The image metadata is stored in image_knowledge_base.json and is loaded into memory for semantic matching.

```
{
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "id": { "type": "string", "description": "Unique filename of the extracted image" },
      "file_path": { "type": "string", "description": "Local path to the image file" },
      "problem_name": { "type": "string", "description": "The specific task or topic identified by the VLM" },
      "dense_caption": { "type": "string", "description": "A detailed technical description of the action in the image" },
      "detected_objects": { "type": "array", "items": { "type": "string" }, "description": "List of components visible" }
    }
  }
}
```

3. Generated Guide Schema (Pipeline Output)

This is the schema of the JSON object returned by the orchestrator and consumed by the React frontend.

```
{
  "type": "object",
  "properties": {
    "status": { "type": "string", "enum": ["success", "error"] },
    "task_title": { "type": "string", "description": "The name of the service procedure" },
    "steps": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "step": { "type": "integer" },
          "instruction": { "type": "string", "description": "The technical text instruction" },
          "chunks": { "type": "array", "items": { "type": "integer" }, "description": "Reference IDs for text sources" },
          "images": {
            "type": ["array", "null"],
            "items": { "type": "string" },
            "description": "List of paths to semantically matched images"
          }
        }
      }
    }
  }
}
```